$$\frac{\text{Number of afflicted persons}}{\text{Number of true and false positives}}$$

here 1 in 51.

Think of the number of times you will be given a medication that carries damaging side effects for a given disease you were told you had, when you may only have 2% probability of being afflicted with it!

WE ARE OPTION BLIND

As an option trader, I have noticed that people tend to undervalue options as they are usually unable to correctly mentally evaluate instruments that deliver an *uncertain* payoff, even when they are fully conscious of the mathematics. Even regulators reinforce such ignorance by explaining to people that options are a *decaying* or *wasting* asset. Options that are out of the money are deemed to *decay*, by losing their premium between two dates.

I will clarify next with a simplified (but sufficient) explanation of what an option means. Say a stock trades at $100 and that someone gives me the right (but not the obligation) to buy it at $110 one month ahead of today. This is dubbed a *call* option. It only makes sense for me to *exercise* it, by asking the seller of the option to deliver me the stock at $110, if it trades at a higher price than $110 in one month's time. If the stock goes to $120, my option will be worth $10, for I will be able to buy the stock at $110 from the option writer and sell it to the market at $120, pocketing the difference. But this does not have a very high probability. It is called *out-of-the-money*, for I have no gain from exercising it right away.

Consider that I buy the option for $1. What do I expect the value of the option to be one month from now? Most people think 0. That is not true. The option has a high probability, say 90% of being worth 0 at expiration, but perhaps 10% probability to be worth an average of $10. Thus, selling the option to me for $1 does not provide the seller with free money. If the sellers had instead bought the stock themselves at $100 and waited the month, they could have sold it for $120. Making $1 now was hardly, therefore, free money. Likewise, buying it is not a

wasting asset. Even professionals can be fooled. How? They confuse the expected value and the most likely scenario (here the expected value is $1 and the most likely scenario is for the option to be worth 0). They mentally overweigh the state that is the most likely, namely, that the market does not move at all. The option is simply the weighted average of the possible states the asset can take.

There is another type of satisfaction provided by the option seller. It is the steady return and the steady feeling of reward – what psychologists call *flow*. It is very pleasant to go to work in the morning with the expectation of being up some small money. It requires some strength of character to accept the expectation of bleeding a little, losing pennies on a steady basis even if the strategy is bound to be profitable over longer periods. I noticed that very few option traders can maintain what I call a "long volatility" position, namely a position that will most likely lose a small quantity of money at expiration, but is expected to make money in the long run because of occasional spurts. I discovered very few people who accepted losing $1 for most expirations and making $10 once in a while, even if the game were fair (i.e., they made the $10 more than 10% of the time).

I divide the community of option traders into two categories: *premium sellers* and *premium buyers*. Premium sellers (also called option sellers) sell options, and generally make steady money, like John in Chapters 1 and 5. Premium buyers do the reverse. Option sellers, it is said, eat like chickens and go to the bathroom like elephants. Alas, most option traders I encountered in my career are *premium sellers* – when they blow up it is generally other people's money.

How could professionals seemingly aware of the (simple) mathematics be put in such a position? Our understanding of math can remain quite superficial; medicine is starting to believe that our actions are not quite guided by the parts of our brain that dictate rationality (see Antonio Damasio's *Descartes' Error* or Ledoux's *Emotional Brain*[11]). We think with our emotions and there is no way around it. For the same reason, people who are otherwise rational engage in smoking or in fights that get them no immediate benefits, likewise people sell options even when they know that it is not a good thing to do. But things can get worse. There is a category of people, generally academics, who, instead of fitting their actions to their brains,